SmartPark Solutions: Project Proposal

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Introduction

Project Objectives

The primary objectives of SmartPark Solutions are to help drivers quickly find available parking, reserve and pay for parking spaces in advance, provide directions to the parking spot, and manage their reservations through a user-friendly web application that also provides notifications. The system will also allow parking lot managers to maintain accurate information about their facilities, including parking availability and pricing. Together, these features aim to reduce the time spent searching for parking and improve the overall parking experience.

Developer Objectives include:

- Apply systems analysis and design techniques to plan a web application.
- Build a dynamic web application using HTML, CSS, Bootstrap, PHP, and related course technologies.
- Use Figma/Lucid, GitHub Codespaces, Visual Studio Code, Git, and GitHub during development.
- Use AI tools such as ChatGPT responsibly for planning, coding, testing, and troubleshooting.
- Collaborate effectively in a software development team.
- Create and present a professional project suitable for a student portfolio.

Description of the Proposed Project

SmartPark Solutions is a web application designed to simplify the process of finding and managing parking spaces for drivers in urban and suburban areas. This app interfaces with sensors or manual entry points at parking facilities to provide real-time information on the availability of parking spots. It's aimed at reducing the time spent searching for parking, thereby cutting down on traffic congestion and emissions.

Motivation

This project was chosen by Group 7 because it is very frustrating to find and manage parking for drivers in urban and suburban areas. It has impactful benefits such as reducing congestion, improved user experience, efficiency of finding parking spots leading to lower emissions and less fuel wastage. Instead of manually looking for parkings, a dynamic website builds the platform to solve the problem. It is a highly demanding and much needed project! A few of our

team members worked on mobile payments, traffic navigation, notification services and advanced booking features in applications before. Therefore, the majority of the team voted for this project.

Objectives and Benefits to the Customer

1. Efficiency: The application reduces the time drivers spend looking for parking, making the process more efficient.
2. Reduced Congestion: It decreases the time cars spend circling for parking, overall traffic congestion is lessened.
3. Environmental Benefits: As a result of reduced time searching for parking, lower emissions are emitted and less fuel wasted.
4. Improved Experience: The app provides easy access to information and payments which enhances the parking experience.
5. Data Collection: It collects valuable data on parking trends which can be used for urban planning and management.

CRUD Use Cases and Roles

Roles

The Two roles used at this stage are the Lot Manager, who maintains the parking lots, and Driver, who finds and reserves parking. Accounts for drivers and payments may be planned for the final submission.

Role	**Who**	**What they can do**
Lot manager (Administrator)	Staff who run a parking facility	Add, edit, and delete parking lots; view all reservations
Driver	Anyone using the app to park	View lots and availability; create, change, and cancel their own reservations

CRUD Set 1: Parking Lot Management

Each lot stores its location, total spots, available spots, and hourly rate. Each use case supports the objective of cutting the time drivers spend searching for parking.

Use case	**Role**	**How it supports the objective**

View (read) lots and real time availability	Driver	Shows open spots in a lot right now, so drivers don't circle looking for parking
Add (create) a parking lot	Lot manager	Puts a new facility into the app so drivers can find it
Update a parking lot	Lot manager	Keeps spot counts and hourly rates current so the availability drivers see is accurate
Delete a parking lot	Lot manager	Removes closed or incorrect lots so drivers aren't sent to the wrong place

CRUD Set 2: Parking Reservation Management

Drivers can reserve an available parking spot in advance and manage their reservations. Each reservation includes the selected parking lot, reservation date, start and end times, and reservation status. This supports the objective of making parking more convenient while reducing the time drivers spend searching for an available spot

Use case	**Role**	**How it supports the objective**
View (read) reservations	Driver / Lot Manager	Allows drivers to check their reservation details and lets lot managers see upcoming reservations to better manage parking availability.
Create a reservation	Driver	Allows drivers to book an available parking spot in advance instead of spending time searching when they arrive.
Update a reservation	Driver	Allows drivers to change their reservation times when plans change, helping keep parking availability accurate.
Delete (cancel) a reservation	Driver	Allows drivers to cancel reservations they no longer

		need, freeing up parking spaces for other drivers.

Team Work and Ethics

Strategies for a Successful Outcome

- Shared GitHub repository: we all work from one repo, each take our own branch, and merge through pull requests so everyone can see what changed and nothing gets overwritten.
- Divide roles and responsibilities for each project based on our strengths and interests of each team member. Hold each other accountable.
- Keep progress with a weekly meeting where we demonstrate what we worked on and resolve issues.

Ethics in Design and Project Management

Since SmartPark Solutions may handle personal information and reservation details, we will design the application so users can only access information relevant to their role. Ensure that we protect the privacy of users and prioritize the needs of users by solving an existing problem instead of creating a problem. We will also be transparent about what information is collected and how it is used.

As a team, we will review and test our work, including any AI-assisted code or content, to ensure that our final submission is accurate, original, and something each contributing member understands. No copy/pasting and producing original work. Overall, we will be transparent and ethical about the items we produce.

For project management, we hold each other accountable. We will establish a team code of conduct and agree on what ethical behavior looks like in our team. We will treat each member fairly. Ensure that we have equal work loads and not sacrifice quality just to meet a deadline.